

SDÜ MÜHENDİSLİK FAKÜLTESİ
BİLGİSAYAR MÜHENDİSLİĞİ BÖLÜMÜ
SAYISAL ÇÖZÜMLEME (NUMERİK ANALİZ) FINAL SINAVI

Soru 1: (a) Aşağıda verilen tablonun son satırını $f'(x)$ için en iyi yaklaşım formüllerini kullanarak doldurunuz.

x	1,8	1,9	2,0	2,1	2,2
f(x)	-1,709847	-1,373823	-1,119214	-0,9160143	-0,7470223
f'(x) = ?					

* $f'(1,8)$ için 5 nokta uç nokta kullanılır. Sebebi ise Hata Oranı 0,2'dir.

$$f'(1,8) = \frac{1}{12h} [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$\frac{1}{12h} [-25f(1,8) + 48f(1,9) - 36f(2) + 16f(2,1) - 3f(2,2)] = \underline{3,899344}$$

* $f'(1,9)$ için 3 nokta orta nokta kullanılmalı. Ama 1,7 bilinmiyor.

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)] \quad h=0,1$$

$$5[f(2,0) - f(1,8)] = \underline{2,953165}$$

* $f'(2,0)$ için 5 nokta orta nokta kullanmalıyız.

$$\frac{1}{12h} [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$\frac{1}{12} [f(1,8) - 8f(1,9) + 8f(2,1) - f(2,2)] = \underline{2,249704}$$

* $f'(2,1)$ için 3 nokta orta nokta kullanılmalı.

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$5[f(2,2) - f(2,0)] = \underline{1,860958}$$

* $f'(2,2)$ için 5 nokta uç nokta kullanılmalı,

$h=-0,1$ için

$$\frac{1}{12h} [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$\frac{1}{12h} [-25f(2,2) + 48f(2,1) - 36f(2,0) + 16f(1,9) - 3f(1,8)] = \underline{1,544209}$$

(b) $f(x) = xe^x$ fonksiyonu için $f''(2)$ türevini aşağıda verilen tabloyu kullanılarak $h=0,1$ ve $h=0,2$ adımları ile yaklaşık hesaplayınız ve mutlak hatayı bulunuz.

x	1,8	1,9	2,0	2,1	2,2
$f(x)$	10,889365	12,703199	14,778112	17,148957	19,855030

* $f'(2)$ ve $h=0,1$ için orta nokta

$$\frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)]$$

$$\frac{1}{(0,1)^2} [f(1,9) - 2f(2) + f(2,1)] = \underline{29,5932}$$

* $f''(2)$ ve $h=0,2$ için orta nokta

$$\frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)]$$

$$\frac{1}{(0,2)^2} [f(1,8) - 2f(2) + f(2,2)] = \underline{29,703525}$$

$$f'(2,0) = 1 \cdot e^x + e^x x = e^x(x+1) =$$

$$f''(2,0) = e^x(x+2) = e^2(2+2) = 29,5562$$

* mutlak hata $h=0,1$ için

$$|29,5562 - 29,5932| = 3,6976 \times 10^{-2}$$

* mutlak hata $h=0,2$ için

$$|29,5562 - 29,7035| = 1,48051 \times 10^{-1}$$

Soru 2:

$e^x + 2^{-x} + 2\cos x - 6 = 0$ denkleminin $[1, 2]$ aralığındaki köklerini Newton-Raphson (Hezetler) yöntemi kullanarak 10^{-4} kesinliği ile bulunuz.

Cevap:

$$f(x) = e^x + 2^{-x} + 2\cos x - 6 = 0$$

$$f'(x) = e^x - (2^{-x} \ln 2) - 2\sin x = 0$$

$x_0 = 1$ alınır.

$n=1$ için:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{f(1)}{f'(1)} \rightarrow 1 - \frac{(1,7011135508)}{(0,68876626856)} = 3,46979801082$$

$$x_1 = 3,46979801052$$

$n=2$ için:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \rightarrow 3,46979801052 - \frac{f(3,46979801052)}{f'(3,46979801052)}$$

$$3,46979801052 - \frac{24,3272653045}{32,7123789904}$$

$$x_2 = 2,7261264699$$

$$\text{Hata payı kontrolü : } |x_2 - x_1| = 0,74367154133 > 10^{-4}$$

işleme devam edilir.

$n=3$ için

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} \Rightarrow 2,72612646919 - \frac{f(2,72612646919)}{f'(2,72612646919)}$$

$$\rightarrow 2,7261264691 - \frac{7,59488418047}{14,361620319}$$

$$x_3 = 2,19729448414$$

Hata payı kontrol: $|x_3 - x_2| = 0,52883198504$ işleme devam edilir.

$n=4$ için

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 2,19729448414 - \frac{f(2,19729448414)}{f'(2,19729448414)}$$

$$\rightarrow 2,19729448414 - \frac{2,06605221491}{7,2293198154}$$

$$x_4 = 1,91427308418$$

Hata payı kontrol $\rightarrow |x_4 - x_3| = 0,28302139996 > 10^{-5}$

işleme devam edilir.

$n=5$ için

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} = 1,91427308418 - \frac{f(1,91427308418)}{f'(1,91427308418)}$$

$$\rightarrow 1,91427308418 - \frac{0,37378703653}{4,71493221038}$$

$$x_5 = 1,83499579665$$

Hata payı kontrol $\rightarrow |x_5 - x_4| = 0,07927728753 > 10^{-4}$

Devam edilir.

$n=6$ için

$$x_6 = x_5 - \frac{f(x_5)}{f'(x_5)} = 1,83499579665 - \frac{f(1,83499579665)}{f'(1,83499579665)}$$

$$\rightarrow 1,83499579665 - \frac{0,02312694970}{4,14022025264}$$

$$x_6 = 1,82940987408$$

Hata payı kontrol $\rightarrow |x_6 - x_5| = 0,005585922 > 10^{-4}$

Devam edilir.

$n=7$ için

$$x_7 = x_6 - \frac{f(x_6)}{f'(x_6)} = 1,82940987408 - \frac{f(1,82940987408)}{f'(1,82940987408)}$$

$$\rightarrow 1,82940987408 - \frac{0,00010975757}{4,10168046543}$$

$$x_7 = 1,82938360251$$

Hata payı kontrol $\rightarrow |x_7 - x_6| = 0,00002627156 < 10^{-4}$

$$\text{Kök} = x_7 = 1,82938360251$$

Soru 3: Dikdörtgen, yamuk ve Simpson metodlarını ayrı ayrı kullanarak $\int_{0.8}^{1.2} \frac{\sin 2x}{x^2} dx$ integralini $n=8$ için yaklaşık hesaplayınız. Sonuçları karşılaştırınız.

Dikdörtgen yöntemi ile;

$$\int_{0.8}^{1.2} \frac{\sin 2x}{x^2} dx \quad n=8 \quad h = \frac{b-a}{n} = \frac{1.2-0.8}{8} = 0.05$$

$$I \cong \int_a^b f(x) dx \cong h [f(0.85) + f(0.9) + f(0.95) + f(1) + f(1.05) + f(1.1) + f(1.15)]$$

$$f(0.85) = 1.3725464504532439$$

$$f(0.9) = 1.2022810257755494$$

$$f(0.95) = 1.0485319531162487$$

$$f(1) = 0.9092974268296817$$

$$f(1.05) = 0.7829563416316314$$

$$f(1.1) = 0.6681788461318925$$

$$f(1.15) = 0.5638602738576332$$

$$I \cong 0.32738261588959405$$

Simpson yöntemi ile

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} dx \quad n=8 \quad h = \frac{b-a}{n} \Rightarrow h = \frac{1,2-0,8}{8} = 0,05$$

$$I \approx f(x)dx \approx \frac{0,05}{3} \left[f(0,8) + 4[f(0,85) + f(0,95) + f(1,05) + f(1,15)] + 2[f(0,9) + f(1) + f(1,1)] + f(1,2) \right]$$

$$f(0,8) = 1,5618337547523515$$

$$f(0,85) = 1,3725464504532434$$

$$f(0,9) = 1,2022810257755492$$

$$f(0,95) = 1,0485319531162482$$

$$f(1) = 0,909297426825681$$

$$f(1,05) = 0,782956341631631$$

$$f(1,1) = 0,6681788461318922$$

$$f(1,15) = 0,5638602738576327$$

$$f(1,2) = 0,4690716531605207$$

$$I \approx 0,37770000136$$

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Yamuk Yöntemi ile

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} \quad n=8 \quad h = \frac{b-a}{n} \Rightarrow h = \frac{1,2 - 0,8}{8} = 0,05$$

$$I \approx \int f(x) dx \approx \frac{h}{2} \left[f(x_0) + 2[f(x_1) + f(x_2) + f(x_3) + f(x_4) + f(x_5) + f(x_6) + f(x_7)] + f(x_8) \right]$$

$$I \approx \frac{0,05}{2} \left[f(0,8) + 2[f(0,85) + f(0,9) + f(0,95) + f(1) + f(1,05) + f(1,1) + f(1,15)] + f(1,2) \right]$$

$$f(0,8) = 1,5618337547523515$$

$$f(0,85) = 1,3725464504532434$$

$$f(0,9) = 1,2022810257755492$$

$$f(1) = 0,909297426825681$$

$$f(1,05) = 0,782956341631631$$

$$f(1,1) = 0,6681788461318922$$

$$f(1,15) = 0,5638602738576327$$

$$f(1,2) = 0,49690716531605207$$

$$I \approx 0,37815546319$$

Soru 41

$u' = xe^{-u} + \frac{x}{1+x^2}$, $u(0) = 1$ başlangıç değer probleminin $[0,1]$ aralığında yaklaşık çözümünü $h=0,025$ adımlı Euler metodu; $h=0,05$ adımlı Heun metodu ve $h=0,1$ adımlı dördüncü dereceden Runge - Kutta metodu ile bulunuz ve sonuçları karşılaştırınız.

Euler metod ile çözüm

$$u_1 = u_0 + hf(x_0, u_0) = 1 + (0.025)f(0,1) = 1 + (0.025) \cdot (0)$$
$$\boxed{= 1}$$

$$u_2 = u_1 + hf(x_1, u_1) = 1 + (0.025)f(0.025, 1) = 1 + (0.025) \cdot (0.03418)$$
$$\boxed{= 1.00085}$$

$$u_3 = u_2 + hf(x_2, u_2) = 1.00085 + (0.025)f(0.05, 1.00085)$$
$$= 1.00085 + (0.025) \cdot (0.06825)$$
$$\boxed{= 1.00256}$$

$$u_4 = u_3 + hf(x_3, u_3) = 1.00256 + (0.025)f(0.075, 1.00256)$$
$$= 1.00256 + (0.025) \cdot (0.1021)$$
$$\boxed{= 1.00511}$$

$$u_5 = u_4 + hf(x_4, u_4) = 1.00511 + (0.025)f(0.1, 1.00511)$$
$$= 1.00511 + (0.025) \cdot (0.13561)$$
$$\boxed{= 1.0085}$$

$$u_6 = u_5 + hf(x_5, u_5) = 1.0085 + (0.025)f(0.125, 1.0085)$$
$$= 1.0085 + (0.025) \cdot (0.16867)$$
$$\boxed{= 1.01272}$$

$$u_7 = u_6 + hf(x_6, u_6) = 1.01272 + (0.025)f(0.15, 1.01272)$$
$$= 1.01272 + (0.025) \cdot (0.20118)$$
$$\boxed{= 1.01775}$$

$$u_8 = u_7 + hf(x_7, u_7) = 1.01775 + (0.025)f(0.175, 1.01775)$$
$$= 1.01775 + (0.025) \cdot (0.2305)$$
$$\boxed{= 1.02358}$$

$$u_9 = u_8 + hf(x_8, u_8) = 1.02358 + (0.025)f(0.2, 1.02358)$$

$$= 1.02358 + (0.025) \cdot (0.26417)$$

$$= 1.03018$$

$$u_{10} = u_9 + hf(x_9, u_9) = 1.03018 + (0.025)f(0.225, 1.03018)$$

$$= 1.03018 + (0.025) \cdot (0.29447)$$

$$= 1.03754$$

$$u_{11} = u_{10} + hf(x_{10}, u_{10}) = 1.03754 + (0.025)f(0.25, 1.03754)$$

$$= 1.03754 + (0.025) \cdot (0.32388)$$

$$= 1.04564$$

$$u_{12} = u_{11} + hf(x_{11}, u_{11}) = 1.04564 + (0.025)f(0.275, 1.04564)$$

$$= 1.04564 + (0.025) \cdot (0.37974)$$

$$= 1.05445$$

$$u_{13} = u_{12} + hf(x_{12}, u_{12}) = 1.05445 + (0.025)f(0.3, 1.05445)$$

$$= 1.05445 + (0.025) \cdot (0.37974)$$

$$= 1.06394$$

$$u_{14} = u_{13} + hf(x_{13}, u_{13}) = 1.06394 + (0.025)f(0.325, 1.06394)$$

$$= 1.06394 + (0.025) \cdot (0.40611)$$

$$= 1.07409$$

$$u_{15} = u_{14} + hf(x_{14}, u_{14}) = 1.07409 + (0.025)f(0.35, 1.07409)$$

$$= 1.07409 + (0.025) \cdot (0.43137)$$

$$= 1.08488$$

$$u_{16} = u_{15} + hf(x_{15}, u_{15}) = 1.08488 + (0.025)f(0.375, 1.08488)$$

$$= 1.08488 + (0.025) \cdot (0.4555)$$

$$= 1.09626$$

$$u_{17} = u_{16} + hf(x_{16}, u_{16}) = 1.09626 + (0.025)f(0.4, 1.09626)$$

$$= 1.09626 + (0.025) \cdot (0.47847)$$

$$= 1.10823$$

$$u_{18} = u_{17} + hf(x_{17}, u_{17}) = 1.10823 + (0.025)f(0.425, 1.10823)$$

$$= 1.10823 + (0.025) \cdot (0.50029)$$

$$= 1.12073$$

$$u_{19} = u_{18} + hf(x_{18}, u_{18}) = 1.12073 + (0.025)f(0.45, 1.12073) \\ = 1.12073 + (0.025) \cdot (0.52094) \\ = 1.13376$$

$$u_{20} = u_{19} + hf(x_{19}, u_{19}) = 1.13376 + (0.025)f(0.475, 1.13376) \\ = 1.13376 + (0.025) \cdot (0.54012) \\ = 1.14727$$

$$u_{21} = u_{20} + hf(x_{20}, u_{20}) = 1.14727 + (0.025)f(0.5, 1.14727) \\ = 1.14727 + (0.025) \cdot (0.55875) \\ = 1.16124$$

$$u_{22} = u_{21} + hf(x_{21}, u_{21}) = 1.16124 + (0.025)f(0.525, 1.16124) \\ = 1.16124 + (0.025) \cdot (0.57594) \\ = 1.17564$$

$$u_{23} = u_{22} + hf(x_{22}, u_{22}) = 1.17564 + (0.025)f(0.55, 1.17564) \\ = 1.17564 + (0.025) \cdot (0.59201) \\ = 1.19044$$

$$u_{24} = u_{23} + hf(x_{23}, u_{23}) = 1.19044 + (0.025)f(0.575, 1.19044) \\ = 1.19044 + (0.025) \cdot (0.60698) \\ = 1.20561$$

$$u_{25} = u_{24} + hf(x_{24}, u_{24}) = 1.20561 + (0.025)f(0.6, 1.20561) \\ = 1.20561 + (0.025) \cdot (0.62088) \\ = 1.22113$$

$$u_{26} = u_{25} + hf(x_{25}, u_{25}) = 1.22113 + (0.025)f(0.625, 1.22113) \\ = 1.22113 + (0.025) \cdot (0.63375) \\ = 1.23698$$

$$u_{27} = u_{26} + hf(x_{26}, u_{26}) = 1.23698 + (0.025)f(0.65, 1.23698) \\ = 1.23698 + (0.025) \cdot (0.64561) \\ = 1.25312$$

$$u_{28} = u_{27} + hf(x_{27}, u_{27}) = 1.25312 + (0.025)f(0.675, 1.25312) \\ = 1.25312 + (0.025) \cdot (0.65651) \\ = 1.26953$$

$$u_{29} = u_{28} + hf(x_{28}, u_{28}) = 1.26953 + (0.025)f(0.7, 1.26953) \\ = 1.26953 + (0.025) \cdot (0.66647) \\ = 1.28619$$

$$u_{30} = u_{29} + hf(x_{29}, u_{29}) = 1.28619 + (0.025)f(0.725, 1.28619) \\ = 1.28619 + (0.025) \cdot (0.67555) \\ = 1.30308$$

$$u_{31} = u_{30} + hf(x_{30}, u_{30}) = 1.30308 + (0.025)f(0.75, 1.30308) \\ = 1.30308 + (0.025) \cdot (0.68377) \\ = 1.32017$$

$$u_{32} = u_{31} + hf(x_{31}, u_{31}) = 1.32017 + (0.025)f(0.775, 1.32017) \\ = 1.32017 + (0.025) \cdot (0.69118) \\ = 1.33745$$

$$u_{33} = u_{32} + hf(x_{32}, u_{32}) = 1.33745 + (0.025)f(0.8, 1.33745) \\ = 1.33745 + (0.025) \cdot (0.69782) \\ = 1.3549$$

$$u_{34} = u_{33} + hf(x_{33}, u_{33}) = 1.3549 + (0.025)f(0.825, 1.3549) \\ = 1.3549 + (0.025) \cdot (0.70372) \\ = 1.37249$$

$$u_{35} = u_{34} + hf(x_{34}, u_{34}) = 1.37249 + (0.025)f(0.85, 1.37249) \\ = 1.37249 + (0.025) \cdot (0.70892) \\ = 1.39021$$

$$u_{36} = u_{35} + hf(x_{35}, u_{35}) = 1.39021 + (0.025)f(0.875, 1.39021) \\ = 1.39021 + (0.025) \cdot (0.71347) \\ = 1.40805$$

$$u_{37} = u_{36} + hf(x_{36}, u_{36}) = 1.40805 + (0.025)f(0.9, 1.40805) \\ = 1.40805 + (0.025) \cdot (0.7174) \\ = 1.42599$$

$$u_{38} = u_{37} + hf(x_{37}, u_{37}) = 1.42599 + (0.025)f(0.925, 1.42599) \\ = 1.42599 + (0.025) \cdot (0.72074) \\ = 1.444$$

$$u_{39} = u_{38} + h_f(x_{38}, u_{38}) = 1.444 + (0.025) f(0.95, 1.444) \\ = 1.444 + (0.025) \cdot (0.72352) \\ = 1.46209$$

$$u_{40} = u_{39} + h_f(x_{39}, u_{39}) = 1.46209 + (0.025) f(0.975, 1.46209) \\ = 1.46209 + (0.025) \cdot (0.7158) \\ = 1.48024$$

$$u(1) = 1.48024$$

4. Dereceden Runge - Kutta Metodu

$$* k_1 = hf(x_0, v_0) = (0.1) f(0, 1) = (0.1) \cdot (0) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, v_0 + \frac{k_1}{2}\right) = (0.1) f(0.05, 1) = (0.1) \cdot (0.06827)$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, v_0 + \frac{k_1}{2}\right) = (0.1) f(0.05, 1.100341) = (0.1) \cdot (0.13555) \\ = 0.00682$$

$$k_4 = hf(x_0 + h, v_0 + k_3) = (0.1) f(0.1, 1.00682) = (0.1) \cdot (0.13555) \\ = 0.01355$$

$$v_1 = v_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$v_1 = 1 + \frac{1}{6} [0 + 2(0.00682) + 2(0.00682) + (0.01355)]$$

$$v_1 = 1.00681$$

$$v(0.1) = 1.00681$$

$$* k_1 = hf(x_1, v_1) = (0.1) f(0.1, 1.00681) = (0.1) \cdot (0.13555) \\ = 0.01355$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, v_1 + \frac{k_1}{2}\right) = (0.1) f(0.15, 1.01359) \\ = (0.1) \cdot (0.20114) \\ = 0.02011$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, v_1 + \frac{k_1}{2}\right) = (0.1) f(0.15, 1.01687) \\ = (0.1) \cdot (0.20096) \\ = 0.0201$$

$$k_4 = hf(x_1 + h, v_1 + k_3) = (0.1) f(0.2, 1.0269) \\ = (0.1) \cdot (0.26393) \\ = 0.02639$$

$$v_2 = v_1 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$v_2 = 1.00681 + \frac{1}{6} [0.01355 + 2(0.02011) + 2(0.0201) + (0.02639)]$$

$$v_2 = 1.02687$$

$$v(0.2) = 1.02687$$

$$k_1 = hf(x_2, u_2) = (0.1) f(0.2, 1.02687) = (0.1) \cdot (0.26393) \\ = 0.02639$$

$$k_2 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_1}{2}\right) = (0.1) f(0.25, 1.04007) \\ = (0.1) (0.3265) \\ = 0.03237$$

$$k_3 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_2}{2}\right) = (0.1) f(0.25, 1.04205) \\ = (0.1) \cdot (0.32339) \\ = 0.03234$$

$$k_4 = hf(x_2 + h, u_2 + k_3) = (0.1) f(0.3, 1.05921) = (0.1) \cdot (0.37925) \\ = 0.03792$$

$$u_3 = u_2 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_3 = 1.02687 + \frac{1}{6} [0.02639 + 2(0.03237) + 2(0.03234) + (0.03792)]$$

$$u_3 = 1.05916$$

$$u(0.3) = 1.05916$$

$$k_1 = hf(x_3, u_3) = (0.1) f(0.3, 1.05916) = (0.1) \cdot (0.37925) \\ = 0.03793$$

$$k_2 = hf\left(x_3 + \frac{h}{2}, u_3 + \frac{k_1}{2}\right) = (0.1) f(0.35, 1.07812) = (0.1) \cdot (0.43089) \\ = 0.04309$$

$$k_3 = hf\left(x_3 + \frac{h}{2}, u_3 + \frac{k_2}{2}\right) = (0.1) f(0.35, 1.0807) = (0.1) \cdot (0.430) \\ = 0.04306$$

$$k_4 = hf(x_3 + h, u_3 + k_3) = (0.1) f(0.4, 1.10222) = (0.1) \cdot (0.47768) \\ = 0.04777$$

$$u_4 = u_3 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_4 = 1.05916 + \frac{1}{6} [0.03793 + 2(0.04309) + 2(0.04306) + (0.04777)]$$

$$u_4 = 1.10215$$

$$u(0.4) = 1.10215$$

$$k_1 = hf(x_4, u_4) = (0.1) f(0.4, 1.0215) = (0.1) \cdot (0.47769) \\ = 0.04777$$

$$k_2 = hf\left(x_4 + \frac{h}{2}, u_4 + \frac{k_1}{2}\right) = (0.1) f(0.45, 1.12604) = (0.1) \cdot (0.52016) \\ = 0.05202$$

$$k_3 = hf\left(x_4 + \frac{h}{2}, u_4 + \frac{k_2}{2}\right) = (0.1) f(0.45, 1.12816) = (0.1) \cdot (0.51985) \\ = 0.05199$$

$$k_4 = hf(x_4 + h, u_4 + k_3) = (0.1) f(0.5, 1.15414) = (0.1) \cdot (0.55766) \\ = 0.05577$$

$$u_5 = u_4 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_5 = 1.10215 + \frac{1}{6} [0.04777 + 2(0.05202) + 2(0.05199) + (0.05577)]$$

$$u_5 = 1.15408 \quad \boxed{u(0.5) = 1.15408}$$

$$k_1 = hf(x_5, u_5) = (0.1) f(0.5, 1.15408) = (0.1) \cdot (0.55764) = 0.05577$$

$$k_2 = hf\left(x_5 + \frac{h}{2}, u_5 + \frac{k_1}{2}\right) = (0.1) f(0.55, 1.18196) = (0.1) \cdot (0.59094) = 0.05909$$

$$k_3 = hf\left(x_5 + \frac{h}{2}, u_5 + \frac{k_2}{2}\right) = (0.1) f(0.55, 1.18362) = (0.1) \cdot 0.59066 = 0.05907$$

$$k_4 = hf(x_5 + h, u_5 + k_3) = (0.1) f(0.6, 1.21316) = (0.1) \cdot (0.61953) = 0.06195$$

$$u_6 = u_5 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_6 = 1.15408 + \frac{1}{6} [0.05577 + 2(0.05909) + 2(0.05907) + (0.06195)] = \boxed{1.21308}$$

$$\boxed{u(0.6) = 1.21308}$$

$$\rightarrow k_1 = hf(x_6, u_6) = (0.1) f(0.6, 1.21308) = (0.1) \cdot (0.6195) = 0.06195$$

$$k_2 = hf\left(x_6 + \frac{h}{2}, u_6 + \frac{k_1}{2}\right) = (0.1) f(0.65, 1.24406) = (0.1) \cdot (0.64428) = 0.0644$$

$$k_3 = hf\left(x_6 + \frac{h}{2}, u_6 + \frac{k_2}{2}\right) = (0.1) f(0.65, 1.2453) = (0.1) \cdot (0.64405) = 0.0644$$

$$k_4 = hf(x_6 + h, u_6 + k_3) = (0.1) f(0.7, 1.27749) = (0.1) \cdot (0.66491) = 0.06649$$

$$u_7 = u_6 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_7 = 1.21308 + \frac{1}{6}[0.06195 + 2(0.06443) + 2(0.0644) + (0.06649)] = 1.27744$$

$$u(0.7) = 1.27744$$

$$\rightarrow k_1 = hf(x_7, u_7) = (0.1) f(0.7, 1.27744) = (0.1) \cdot (0.66492) = 0.06649$$

$$k_2 = hf\left(x_7 + \frac{h}{2}, u_7 + \frac{k_1}{2}\right) = (0.1) f(0.75, 1.31068) = (0.1) \cdot (0.68223) = 0.06822$$

$$k_3 = hf\left(x_7 + \frac{h}{2}, u_7 + \frac{k_2}{2}\right) = (0.1) f(0.75, 1.3155) = (0.1) \cdot (0.68205) = 0.06821$$

$$k_4 = hf(x_7 + h, u_7 + k_3) = (0.1) f(0.8, 1.34564) = (0.1) \cdot (0.6961) = 0.06961$$

$$u_8 = u_7 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_8 = 1.27744 + \frac{1}{6}[0.06649 + 2(0.06822) + 2(0.06821) + (0.06961)] = 1.3456$$

$$u(0.8) = 1.3456$$

$$\begin{aligned}
 k_1 &= hf(x_8, v_8) = (0.1) \cdot f(0.8, 1.3456) = (0.1) \cdot (0.69611) = 0.06961 \\
 k_2 &= hf\left(x_8 + \frac{h}{2}, v_8 + \frac{k_1}{2}\right) = (0.1) \cdot f(0.85, 1.3804) = (0.1) \cdot (0.70722) = 0.07072 \\
 k_3 &= hf\left(x_8 + \frac{h}{2}, v_8 + \frac{k_2}{2}\right) = (0.1) \cdot f(0.85, 1.38096) = (0.1) \cdot (0.70711) = 0.07071 \\
 k_4 &= hf(x_8 + h, v_8 + k_3) = (0.1) \cdot f(0.9, 1.41631) = (0.1) \cdot (0.71559) = 0.07156 \\
 v_9 &= v_8 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\
 v_9 &= 1.3456 + \frac{1}{6}[0.06961 + 2(0.07072) + 2(0.07071) + (0.07156)] = 1.41627 \\
 \boxed{v(0.9) = 1.41627}
 \end{aligned}$$

$$\begin{aligned}
 k_1 &= hf(x_9, v_9) = (0.1) \cdot f(0.9, 1.41627) = (0.1) \cdot (0.71559) = 0.07156 \\
 k_2 &= hf\left(x_9 + \frac{h}{2}, v_9 + \frac{k_1}{2}\right) = (0.1) \cdot f(0.95, 1.45205) = (0.1) \cdot (0.72173) = 0.07217 \\
 k_3 &= hf\left(x_9 + \frac{h}{2}, v_9 + \frac{k_2}{2}\right) = (0.1) \cdot f(0.95, 1.45236) = (0.1) \cdot (0.72166) = 0.07216 \\
 k_4 &= hf(x_9 + h, v_9 + k_3) = (0.1) \cdot f(1, 1.48843) = (0.1) \cdot (0.72573) = 0.07257 \\
 v_{10} &= v_9 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\
 v_{10} &= 1.41627 + \frac{1}{6}[0.07156 + 2(0.07217) + 2(0.07216) + (0.07257)] = 1.4884 \\
 \boxed{v(1) = 1.4884}
 \end{aligned}$$

Heun Metodu

$$u_{n+1} = u_n + \frac{h}{2} (f(x_n, u_n) + f(x_{n+1}, \tilde{u}_{n+1}))$$

1. Adım

$$x_1 = x_0 + h = 0 + \frac{1}{20} = \frac{1}{20}$$

$$\tilde{u}_1 = u_0 + hf(x_0, u_0) = 1 + hf(0, 1) = (1) + \frac{1}{20} (0) = 1$$

$$\begin{aligned} u_1 &= u_0 + \frac{h}{2} (f(x_0, u_0) + f(x_1, \tilde{u}_1)) = 1 + \frac{h}{2} (f(0, 1) + f(\frac{1}{20}, 1)) \\ &= 1 + \frac{\frac{1}{20}}{2} (0 + (\frac{1}{20} + \frac{20}{401})) = \boxed{1.00170673209448} \end{aligned}$$

2. Adım

$$x_2 = x_1 + h = \frac{1}{20} + \frac{1}{20} = \frac{1}{10}$$

$$\begin{aligned} \tilde{u}_2 &= u_1 + hf(x_1, u_1) = 1.00170673209448 + h \cdot f(\frac{1}{20}, 1.00170673209448) \\ &= (1.00170673209448) + \frac{1}{20} \cdot (0.0682379169717956) = 1.00511862794307 \end{aligned}$$

$$\begin{aligned} u_2 &= u_1 + \frac{h}{2} (f(x_1, u_1) + f(x_2, \tilde{u}_2)) = 1.00170673209448 + \frac{h}{2} (f(\frac{1}{20}, 1.00170673209448) \\ &+ f(\frac{1}{10}, 1.00511862794307)) = 1.00170673209448 + \frac{\frac{1}{20}}{2} (0.0682379169717956 \\ &+ (0.13561002241584)) = \boxed{1.00680293057917} \end{aligned}$$

3. Adım

$$x_3 = x_2 + h = \frac{1}{10} + \frac{1}{20} = \frac{3}{20}$$

$$\tilde{u}_3 = u_2 + h \cdot f(x_2, y_2) = 1.00680293057917 + h \cdot f\left(\frac{1}{10}, 1.00680293057917\right) =$$

$$= (1.00680293057917) + \frac{1}{20} \cdot (0.135548428620698) = 1.01358035201021$$

$$u_3 = u_2 + \frac{h}{2} f(x_2, u_2) + f(x_3, \tilde{u}_3) = 1.00680293057917 +$$

$$+ \frac{h}{2} \left(f\left(\frac{1}{10}, 1.00680293057917\right) + f\left(\frac{3}{20}, 1.01358035201021\right) \right) = 1.00680293057917 +$$

$$+ \frac{1}{20} (0.135548428620698 + (0.201136858365574)) = \boxed{1.01522006275383}$$

4. Adım

$$x_4 = x_3 + h = \frac{3}{20} + \frac{1}{20} = \frac{1}{5}$$

$$\tilde{u}_4 = u_3 + h \cdot f(x_3, u_3) = 1.01522006275383 + h \cdot f\left(\frac{3}{20}, 1.01522006275383\right) =$$

$$= (1.01522006275383) + \frac{1}{20} \cdot (0.201004766960321) = 1.02527244623399$$

$$u_4 = u_3 + \frac{h}{2} (f(x_3, u_3) + f(x_4, \tilde{u}_4)) = 1.01522006275383 + \frac{h}{2} \left(f\left(\frac{3}{20}, 1.01522006275383\right) +$$

$$+ f\left(\frac{1}{5}, 1.02527244623399\right) \right) = 1.01522006275383 + \frac{1}{20} (0.201004766960321)$$

$$+ (0.266047437502724)) = \boxed{1.02684744043148}$$

5. Adım

$$x_5 = x_4 + h = \frac{1}{5} + \frac{1}{20} = \frac{1}{4}$$

$$\tilde{u}_5 = u_4 + h \cdot f(x_4, u_4) = 1.02684744043148 + h \cdot f\left(\frac{1}{5}, 1.02684744043148\right) =$$

$$= (1.02684744043148) + \frac{1}{20} \cdot (0.263934536752664) = 1.040004416726911$$

$$u_5 = u_4 + \frac{h}{2} (f(x_4, u_4) + f(x_5, \tilde{u}_5)) = 1.02684744043148 + \frac{h}{2} \left(f\left(\frac{1}{5}, 1.02684744043148\right) +$$

$$+ f\left(\frac{1}{4}, 1.040004416726911\right) \right) = 1.02684744043148 + \frac{1}{20} (0.263934536752664 +$$

$$+ (0.323653885440926)) = \boxed{1.04153715098632}$$

6. Adım

$$x_6 = x_5 + h = \frac{1}{4} + \frac{1}{20} = \frac{3}{10}$$

$$\begin{aligned}\tilde{u}_6 &= u_5 + h \cdot f(x_5, u_5) = 1.04153715098632 + h \cdot f\left(\frac{1}{4}, 1.04153715098632\right) = \\ &= (1.04153715098632) + \frac{1}{20} \cdot (0.323522064174304) = 1.05771325419503 \\ u_6 &= u_5 + \frac{h}{2} (f(x_5, u_5) + f(x_6, \tilde{u}_6)) = 1.04153715098632 + \frac{h}{2} \left(f\left(\frac{1}{4}, 1.04153715098632\right) + \right. \\ &\left. f\left(\frac{3}{10}, 1.05771325419503\right) \right) = 1.04153715098632 + \frac{1}{20} (0.323522064174304 + 0.3796040697690) \\ &= 1.0591103038349\end{aligned}$$

7. Adım

$$x_7 = x_6 + h = \frac{3}{10} + \frac{1}{20} = \frac{7}{20}$$

$$\begin{aligned}\tilde{u}_7 &= u_6 + h \cdot f(x_6, u_6) = 1.0591103038349 + h \cdot f\left(\frac{3}{10}, 1.0591103038349\right) = \\ &= (1.0591103038349) + \frac{1}{20} \cdot (0.379258614167179) = 1.07807323454326\end{aligned}$$

$$\begin{aligned}u_7 &= u_6 + \frac{h}{2} (f(x_6, u_6) + f(x_7, \tilde{u}_7)) = 1.0591103038349 \\ &+ \frac{h}{2} \left(f\left(\frac{3}{10}, 1.0591103038349\right) + f\left(\frac{7}{20}, 1.07807323454326\right) \right) = 1.0591103038349 \\ &+ \frac{1}{20} (0.379258614167179 + 0.430891675977526) = 1.07936406108852\end{aligned}$$

8. Adım

$$x_8 = x_7 + h = \frac{7}{20} + \frac{1}{20} = \frac{2}{5}$$

$$\begin{aligned}\tilde{u}_8 &= u_7 + h \cdot f(x_7, u_7) = 1.07936406108852 + h \cdot f\left(\frac{7}{20}, 1.07936406108852\right) = \\ &= (1.07936406108852) + \frac{1}{20} \cdot (0.450738053626896) = 1.10090096376986 \\ u_8 &= u_7 + \frac{h}{2} (f(x_7, u_7) + f(x_8, \tilde{u}_8)) = 1.07936406108852 + \frac{h}{2} \left(f\left(\frac{7}{20}, 1.07936406108852\right) + \right. \\ &\left. f\left(\frac{2}{5}, 1.10090096376986\right) \right) = 1.07936406108852 + \frac{1}{20} (0.430738053626896 + \\ &+ 0.447856111795993) = 1.10207891522409\end{aligned}$$

9. Adım

$$x_9 = x_8 + h = \frac{2}{5} + \frac{1}{20} = \frac{9}{20}$$

$$\tilde{u}_9 = u_8 + h \cdot f(x_8, u_8) = 1.10207891522409 + h \cdot f\left(\frac{2}{5}, 1.10207891522409\right) = \\ = (1.10207891522409) + \frac{1}{20} \cdot (0.477699502907764) = 1.12596387036948$$

$$u_9 = u_8 + \frac{h}{2} (f(x_8, u_8) + f(x_9, \tilde{u}_9)) = 1.10207891522409 + \\ + \frac{h}{2} (f\left(\frac{2}{5}, 1.10207891522409\right) + f\left(\frac{9}{20}, 1.12596387036948\right)) = \\ = 1.10207891522409 + \frac{1}{20} (0.477699502907764 + (0.520173234152475)) \\ = 1.1270257336506$$

10. Adım

$$x_{10} = x_9 + h = \frac{9}{20} + \frac{1}{20} = \frac{1}{2}$$

$$\tilde{u}_{10} = u_9 + h \cdot f(x_9, u_9) = 1.1270257336506 + h \cdot f\left(\frac{9}{20}, 1.1270257336506\right) = (1.1270257336506) + \\ + \frac{1}{20} \cdot (0.520018337341419) = 1.15302665051767$$

$$u_{10} = u_9 + \frac{h}{2} (f(x_9, u_9) + f(x_{10}, \tilde{u}_{10})) = 1.1270257336506 + \frac{h}{2} (f\left(\frac{9}{20}, 1.1270257336506\right) + \\ + f\left(\frac{1}{2}, 1.15302665051767\right)) = 1.1270257336506 + \frac{1}{20} (0.520018337341419 + \\ + (0.557839934684269)) = 1.15397219045124$$

11. Adım

$$x_{11} = x_{10} + h = \frac{1}{2} + \frac{1}{20} = \frac{11}{20}$$

$$\tilde{u}_{11} = u_{10} + h \cdot f(x_{10}, u_{10}) = 1.15397219045124 + h \cdot f\left(\frac{1}{2}, 1.15397219045124\right) = \\ = (1.15397219045124) + \frac{1}{20} \cdot (0.557650761258742) = 1.18185672851418$$

$$u_{11} = u_{10} + \frac{h}{2} (f(x_{10}, u_{10}) + f(x_{11}, \tilde{u}_{11})) = 1.15397219045124 + \\ + \frac{h}{2} (f\left(\frac{1}{2}, 1.15397219045124\right) + f\left(\frac{11}{20}, 1.18185672851418\right)) = 1.15397219045124 + \\ + \frac{1}{20} (0.557650761258742 + (0.590954679347108)) = 1.18268832646639$$

12. Adım

$$x_{12} = x_{11} + h = \frac{11}{20} + \frac{1}{20} = \frac{3}{5}$$

$$\begin{aligned}\tilde{y}_{12} &= y_{11} + h \cdot f(x_{11}, y_{11}) = 1.18268832646639 + h \cdot f\left(\frac{11}{20}, 1.18268832646639\right) = \\ &= (1.18268832646639) + \frac{1}{20} \cdot (0.5308144555644) = 1.21222904924461\end{aligned}$$

$$\begin{aligned}y_{12} &= y_{11} + \frac{h}{2} (f(x_{11}, y_{11}) + f(x_{12}, \tilde{y}_{12})) = 1.18268832646639 \\ &+ \frac{h}{2} (f\left(\frac{11}{20}, 1.18268832646639\right) + f\left(\frac{3}{5}, 1.21222904924461\right)) = \\ &= 1.18268832646639 + \frac{1}{20} (0.5308144555644 + (0.619696064556128)) \\ &= 1.2129510994694\end{aligned}$$

13. Adım

$$x_{13} = x_{12} + h = \frac{3}{5} + \frac{1}{20} = \frac{13}{20}$$

$$\begin{aligned}\tilde{y}_{13} &= y_{12} + h \cdot f(x_{12}, y_{12}) = 1.2129510994694 + h \cdot f\left(\frac{3}{5}, 1.2129510994694\right) = \\ &= (1.2129510994694) + \frac{1}{20} \cdot (0.619567610679437) = 1.2439294800337\end{aligned}$$

$$\begin{aligned}y_{13} &= y_{12} + \frac{h}{2} (f(x_{12}, y_{12}) + f(x_{13}, \tilde{y}_{13})) = 1.2129510994694 + \frac{h}{2} (f\left(\frac{3}{5}, 1.2129510994694\right) \\ &+ f\left(\frac{13}{20}, 1.2439294800337\right)) = 1.2129510994694 + \frac{1}{20} (0.619567610679437 + \\ &+ (0.644304061308033)) = 1.26454789126909\end{aligned}$$

14. Adım

$$x_{14} = x_{13} + h = \frac{13}{20} + \frac{1}{20} = \frac{7}{10}$$

$$\begin{aligned}\tilde{y}_{14} &= y_{13} + h \cdot f(x_{13}, y_{13}) = 1.24454789126909 + h \cdot f\left(\frac{13}{20}, 1.24454789126909\right) = \\ &= (1.24454789126909) + \frac{1}{20} \cdot (0.644188230320012) = 1.27675730278509\end{aligned}$$

$$\begin{aligned}y_{14} &= y_{13} + \frac{h}{2} (f(x_{13}, y_{13}) + f(x_{14}, \tilde{y}_{14})) = 1.24454789126909 \\ &+ \frac{h}{2} (f\left(\frac{13}{20}, 1.24454789126909\right) + f\left(\frac{7}{10}, 1.27675730278509\right)) = \\ &= 1.24454789126909 + \frac{1}{20} (0.644188230320012 + (0.665056005943233)) \\ &= 1.27727901567567\end{aligned}$$

15. Adım

$$x_{15} = x_{14} + h = \frac{7}{10} + \frac{1}{20} = \frac{3}{2}$$

$$\begin{aligned} \tilde{u}_{15} &= u_{14} + h \cdot f(x_{14}, u_{14}) = 1.27727901967567 + h \cdot f\left(\frac{7}{10}, 1.27727901967567\right) = \\ &= (1.27727901967567) + \frac{1}{20} \cdot (0.664955062986014) = 1.31052677282497 \end{aligned}$$

$$\begin{aligned} u_{15} &= u_{14} + \frac{h}{2} (f(x_{14}, u_{14}) + f(x_{15}, \tilde{u}_{15})) = 1.27727901967567 + \\ &+ \frac{h}{2} (f\left(\frac{7}{10}, 1.27727901967567\right) + f\left(\frac{3}{2}, 1.31052677282497\right)) = 1.27727901967567 + \\ &\frac{1}{20} (0.664955062986014 + (0.682258469955683)) = \boxed{1.31095935799921} \end{aligned}$$

16. Adım

$$x_{16} = x_{15} + h = \frac{3}{2} + \frac{1}{20} = \frac{6}{5}$$

$$\begin{aligned} \tilde{u}_{16} &= u_{15} + h \cdot f(x_{15}, u_{15}) = 1.31095935799921 + h \cdot f\left(\frac{3}{2}, 1.31095935799921\right) = \\ &= (1.31095935799921) + \frac{1}{20} \cdot (0.682170994861793) = 1.3450679077423 \end{aligned}$$

$$\begin{aligned} u_{16} &= u_{15} + \frac{h}{2} (f(x_{15}, u_{15}) + f(x_{16}, \tilde{u}_{16})) = 1.31095935799921 + \\ &+ \frac{h}{2} (f\left(\frac{3}{2}, 1.31095935799921\right) + f\left(\frac{6}{5}, 1.3450679077423\right)) = 1.31095935799921 + \\ &+ \frac{1}{20} (0.682170994861793 + (0.696222490686665)) = \boxed{1.34541919513792} \end{aligned}$$

17. Adım

$$x_{17} = x_{16} + h = \frac{6}{5} + \frac{1}{20} = \frac{17}{10}$$

$$\begin{aligned} \tilde{u}_{17} &= u_{16} + h \cdot f(x_{16}, u_{16}) = 1.34541919513792 + h \cdot f\left(\frac{6}{5}, 1.34541919513792\right) = \\ &= (1.34541919513792) + \frac{1}{20} \cdot (0.696149289064476) = 1.38022665959115 \end{aligned}$$

$$\begin{aligned} u_{17} &= u_{16} + \frac{h}{2} (f(x_{16}, u_{16}) + f(x_{17}, \tilde{u}_{17})) = 1.34541919513792 + \\ &+ \frac{h}{2} (f\left(\frac{6}{5}, 1.34541919513792\right) + f\left(\frac{17}{10}, 1.38022665959115\right)) = \\ &= 1.34541919513792 + \frac{1}{20} (0.696149289064476 + (0.707262101660799)) \end{aligned}$$

$$= 1.38050447990605$$

18. Adım

$$x_{18} = x_{17} + h = \frac{17}{20} + \frac{1}{20} = \frac{9}{10}$$

$$\begin{aligned} \tilde{y}_{18} &= y_{17} + h \cdot f(x_{17}, y_{17}) = 1.38050447990605 + h \cdot f\left(\frac{17}{20}, 1.38050447990605\right) = \\ &= (1.38050447990605) + \frac{1}{20} \cdot (0.707202713787077) = 1.41586461559541 \\ y_{18} &= y_{17} + \frac{h}{2} (f(x_{17}, y_{17}) + f(x_{18}, \tilde{y}_{18})) = 1.38050447990605 + \\ &+ \frac{h}{2} (f\left(\frac{17}{20}, 1.38050447990605\right) + f\left(\frac{9}{10}, 1.41586461559541\right)) = \\ &= 1.38050447990605 + \frac{1}{20} (0.707202713787077 + (0.71568166931525)) \\ &= 1.41607658948361 \end{aligned}$$

19. Adım

$$x_{19} = x_{18} + h = \frac{9}{10} + \frac{1}{20} = \frac{19}{20}$$

$$\begin{aligned} \tilde{y}_{19} &= y_{18} + h \cdot f(x_{18}, y_{18}) = 1.41607658948361 + h \cdot f\left(\frac{9}{10}, 1.41607658948361\right) = \\ &= (1.41607658948361) + \frac{1}{20} \cdot (0.71563536977284) = 1.45185835797248 \\ y_{19} &= y_{18} + \frac{h}{2} (f(x_{18}, y_{18}) + f(x_{19}, \tilde{y}_{19})) = 1.41607658948361 + \\ &+ \frac{h}{2} (f\left(\frac{9}{10}, 1.41607658948361\right) + f\left(\frac{19}{20}, 1.45185835797248\right)) = 1.41607658948361 + \\ &+ \frac{1}{20} (0.71563536977284 + (0.721771008232136)) = 1.45201174893385 \end{aligned}$$

20. Adım

$$x_{20} = x_{19} + h = \frac{19}{20} + \frac{1}{20} = 1$$

$$\begin{aligned} \tilde{y}_{20} &= y_{19} + h \cdot f(x_{19}, y_{19}) = 1.45201174893385 + h \cdot f\left(\frac{19}{20}, 1.45201174893385\right) = \\ &= (1.45201174893385) + \frac{1}{20} \cdot (0.721736892398078) = 1.48809859355375 \\ y_{20} &= y_{19} + \frac{h}{2} (f(x_{19}, y_{19}) + f(x_{20}, \tilde{y}_{20})) = 1.45201174893385 + \\ &+ \frac{h}{2} (f\left(\frac{19}{20}, 1.45201174893385\right) + f(1, 1.48809859355375)) = 1.45201174893385 + \\ &\frac{1}{20} (0.721736892398078 + (0.72801588212941)) = 1.48820021094925 \end{aligned}$$

$$y(1) = 1.48820021094925$$

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$$\begin{cases} 4x_1 - x_2 + 2x_3 = 0 \\ -x_1 + 6x_2 - x_3 = 1 \\ 2x_1 - x_2 + 5x_3 = -8 \end{cases}$$

cebirsel lineer denklemler sisteminin yaklaşık çözümünü Gauss-Seidel yöntemi ve 0,001 kesinliği ile bulunuz.

1. yaklaşım

$$x_1 = \frac{1}{4} [0 + (0) - 2(0)] = \frac{1}{4} [0] = 0$$

$$y_1 = \frac{1}{6} [1 + (0) + (0)] = \frac{1}{6} [1] = 0,16667$$

$$z_1 = \frac{1}{5} [-8 - 2(0) + (0,16667)] = \frac{1}{5} [-7,8333] = -1,56667$$

2. yaklaşım

$$x_2 = \frac{1}{4} [0 + (0,16667) - 2(-1,56667)] = \frac{1}{4} [3,3] = 0,825$$

$$y_2 = \frac{1}{6} [1 + (0,825) + (-1,56667)] = \frac{1}{6} [0,25833] = 0,04306$$

$$z_2 = \frac{1}{5} [-8 - 2(0,825) + (0,04306)] = \frac{1}{5} [-9,60694] = -1,92139$$

3. yaklaşım

$$x_3 = \frac{1}{4} [0 + (0,04306) - 2(-1,92139)] = \frac{1}{4} [3,88583] = 0,97146$$

$$y_3 = \frac{1}{6} [1 + (0,97146) + (-1,92139)] = \frac{1}{6} [0,05007] = 0,00834$$

$$z_3 = \frac{1}{5} [-8 - 2(0,97146) + (0,00834)] = \frac{1}{5} [-9,93457] = -1,98691$$

4. yaklaşım

$$x_4 = \frac{1}{4} [0 + (0,00834) - 2(-1,98691)] = \frac{1}{4} [3,98217] = 0,99554$$

$$y_4 = \frac{1}{6} [1 + (0,99554) + (-1,98691)] = \frac{1}{6} [0,00863] = 0,00144$$

$$z_4 = \frac{1}{5} [-8 - 2(0,99554) + (0,00144)] = \frac{1}{5} [-9,98965] = -1,99793$$

5. Yaklaşım

$$x_5 = \frac{1}{4} [0 + (0,00144) - 2(-1,99793)] = \frac{1}{4} [3,9973] = 0,99932$$

$$y_5 = \frac{1}{6} [1 + (0,99932) + (-1,99793)] = \frac{1}{6} [0,00139] = 0,00023$$

$$z_5 = \frac{1}{5} [-8 - 2(0,99932) + (0,00023)] = \frac{1}{5} [-9,99842] = -1,99968$$

6. Yaklaşım

$$x_6 = \frac{1}{4} [0 + (0,00023) - 2(-1,99968)] = \frac{1}{4} [3,9996] = 0,9999$$

$$y_6 = \frac{1}{6} [1 + (0,9999) + (-1,99968)] = \frac{1}{6} [0,00022] = 0,00004$$

$$z_6 = \frac{1}{5} [-8 - 2(0,9999) + (0,00004)] = \frac{1}{5} [-9,99976] = -1,99995$$

$$x_1 = 0,9999 \approx 1$$

$$x_2 = 0,00004 \approx 0$$

$$x_3 = -1,99995 \approx -2$$

$$\text{Hata payları } |x_6 - x_5| = 0,00085$$

$$|y_6 - y_5| = 0,00019$$

$$|z_6 - z_5| = 0,00027$$

Hepsi 0,001'den küçük tir.